

How do different forms of Dynamic Difficulty Adjustment most effectively sustain player flow?

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1 Abstract

Dynamic Difficulty Adjustment (DDA) is a key mechanism for maintaining player engagement by dynamically balancing challenge and ability to sustain the psychological state of Flow. Building upon the theories of Csikszentmihalyi [Csikszentmihalyi, 1990] and Jenova Chen [Chen, 2007], this paper examines how different forms of DDA influence player immersion and flow continuity. Through a comparative analysis of representative games across genres, it identifies the limitations of traditional binary classifications—system-driven versus player-driven—and introduces a refined four-quadrant framework based on two additional dimensions: directionality (positive versus negative) and visibility (implicit versus explicit).

2 Introduction

In digital games, the balance between challenge and ability lies at the core of maintaining player engagement. According to Csikszentmihalyi’s theory of *Flow* [Csikszentmihalyi, 1990], when the difficulty of a task matches the individual’s skill level, players enter a state of deep concentration and immersion, achieving an optimal experience. Building upon this foundation, Jenova Chen [Chen, 2007] further introduced Flow theory into the field of interactive media, emphasizing that video games, unlike static art forms, can dynamically adjust the level of challenge to keep players continuously within the “flow channel.” To achieve such balance, games must integrate clear goals, immediate feedback, and dynamically balanced challenges.

Dynamic Difficulty Adjustment (DDA) is the most direct mechanism to materialize this theory within game systems. Unlike traditional static difficulty selection, DDA dynamically modifies game parameters based on players’ real-time performance, preventing frustration caused by excessive challenge or boredom caused by insufficient challenge [Arey and Wells, 2001]. Essentially, DDA serves as a closed feedback loop between

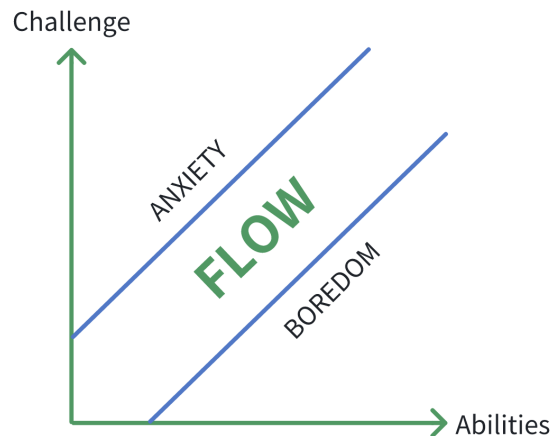


Figure 1: The Flow Channel Model

player behavior and system response: the game continuously monitors player performance data—such as hit accuracy, number of injuries, or completion time—and adjusts enemy strength, resource drop rate, or event pacing in real time to maintain the balance between challenge and ability. However, as Chen [Chen, 2007] emphasized, DDA alone cannot automatically generate Flow; it must work in conjunction with clear goals and immediate feedback mechanisms to truly sustain player immersion. When effectively implemented, DDA transforms “difficulty” from a fixed barrier into a dynamic motivational system, allowing players to remain in a continuous state of Flow where focus, control, and satisfaction intertwine.

The game that truly brought Dynamic Difficulty Adjustment (DDA) into the public eye and gained widespread recognition was *Resident Evil 4* (2005), developed by Capcom. One of its most innovative design features was a hidden system that dynamically adjusted the level of difficulty based on player performance. The game continuously monitored player actions and behavior: if the player skillfully dodged enemy attacks and demonstrated accurate shooting, the system would increase the overall challenge—making enemies more ag-

gressive, raising their damage output, and reducing resource drops. Conversely, if the player frequently took damage or died, the system would lower enemy numbers, slow their reactions, or even decrease enemy encounters altogether to prevent excessive frustration.

This mechanism ensured that players remained within the *flow channel* throughout gameplay—never bored by lack of challenge, nor overwhelmed by excessive difficulty. Importantly, these adjustments occurred entirely in the background, such that most players were unaware of the system’s existence even after completing the game. This subtle and sophisticated control of difficulty allowed *Resident Evil 4* to achieve a delicate balance between tension and playability, establishing it as a landmark example in the application of dynamic difficulty systems.

3 Background

3.1 Operant Conditioning

In the realm of operant conditioning pioneered by B. F. Skinner, a crucial concept is that reinforcement and punishment [Skinner, 2019] in modifying operant behavior serve as excellent tools for thoughtfully guiding player actions. Both reinforcement and punishment are behavioral influences that occur after a stimulus is received by a person or animal. In this context, reinforcement means increasing a behavior following a stimulus, while punishment, in contrast, is about decreasing a behavior. Each can be either positive or negative. Positive indicates adding stimulation, whereas negative is a removal action. By incorporating the perspective of behavioral psychology, we can gain a clearer understanding of experiential preferences for analysis. This allows for a clearer analysis of how changes in game difficulty affect player experience.

3.2 Dynamic Difficulty Adjustment

Academic research on Dynamic Difficulty Adjustment (DDA) has evolved through several distinct stages, each addressing different aspects of adaptive challenge in games. The theoretical foundation was laid by Jenova Chen, who in *Flow in Games* (2007) explicitly connected DDA to Flow theory, framing it as an idealized mechanism for sustaining player engagement. However, Chen emphasized that DDA alone cannot guarantee Flow; it must operate in conjunction with clear goals and immediate feedback to preserve immersion [Chen, 2007]. Following this, Ang and Mitchell [Ang and Mitchell, 2017] conducted one of the first empirical comparisons between system-driven (rDDA) and

player-driven (pDDA) approaches. Their findings revealed that while system-driven adjustment tends to sustain immersion by keeping players unaware of time, player-driven adjustment enhances autonomy and fairness but also increases self-consciousness, interrupting Flow.

Further reviews expanded the conceptual understanding of DDA. Zohaib [Zohaib, 2018] provided a comprehensive survey of existing adaptive systems, identifying player perception as the central challenge of DDA design. When adjustments are too explicit, players feel manipulated; when completely hidden, players may doubt fairness. This “transparency dilemma” remains a core issue in balancing adaptation with immersion. Vang [Vang, 2018] supplemented these findings with experimental data, showing that different DDA implementations influence different dimensions of player experience—such as satisfaction, immersion, and perceived fairness. In parallel, Elias and Salaimeh [Elias and Salaimeh, 2021] extended the discussion to inclusivity, emphasizing that adaptive difficulty should not only enhance Flow but also improve accessibility for players across varying skill and ability levels.

Despite these contributions, several theoretical and practical gaps remain. Most prior studies rely on a binary classification of DDA—system-driven versus player-driven—without considering other critical dimensions such as directionality (positive versus negative) and visibility (implicit versus explicit). This limitation obscures the nuanced ways in which different DDA forms influence Flow across genres. Furthermore, most existing empirical studies focus on a single game genre, such as action games, while cross-genre comparative research remains scarce.

4 Typology and Analysis of Dynamic Difficulty Adjustment (DDA)

To address this gap, and following a comparative analysis of several representative games, this paper proposes a four-quadrant framework for Dynamic Difficulty Adjustment (DDA). The framework aims to provide a more comprehensive understanding of how adaptive difficulty mechanisms operate across different game genres and design philosophies.

Compared with the traditional binary classification of “system-driven vs. player-driven” (Ang & Mitchell, 2017), this model introduces two additional analytical dimensions: **directionality** (positive / negative) and **visibility** (implicit / explicit). The intersection of these two dimensions yields four categories of DDA: *system-driven positive*, *system-driven negative*, *player-driven implicit*, and *player-driven explicit*.

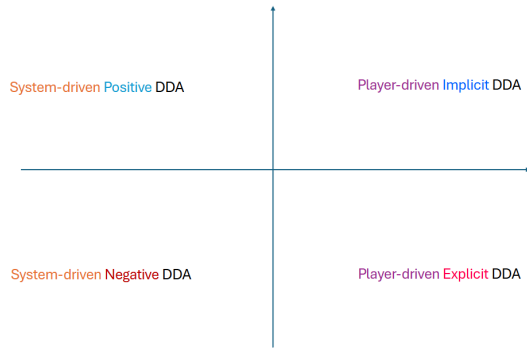


Figure 2: The Four-quadrant Framework

This expanded taxonomy goes beyond conventional technical distinctions, offering a more nuanced tool for analyzing how different DDA strategies influence player immersion, perceived fairness, and flow maintenance. At the same time, it provides a structured conceptual language for cross-genre game design, helping designers determine when and to what extent dynamic difficulty adjustment should be applied in different contexts.

4.1 System-driven Positive DDA

System-driven Positive Dynamic Difficulty Adjustment (DDA) refers to an adaptive mechanism that is entirely controlled by the game system. The system continuously monitors the player’s performance—such as hit accuracy, reaction speed, or number of failures—and automatically adjusts gameplay variables including enemy aggression, resource drop frequency, or encounter intensity. Its primary objective is to dynamically maintain a balance between “challenge and ability,” thereby sustaining the player’s state of flow. When players repeatedly fail or fall into difficulty, the system slightly lowers the challenge to prevent frustration and disengagement; when players perform well, it subtly increases the challenge to maintain tension and engagement. Since these adjustments usually occur in the background, players are often unaware of the changes taking place.

In practice, system-driven Positive DDA is the most widely discussed form of dynamic difficulty both in academic research and in the gaming community. It represents the mainstream understanding of DDA as an implicit assistance mechanism that automatically aligns the challenge level with the player’s capabilities, keeping them consistently within the “flow channel” where concentration, control, and satisfaction are balanced [Csikszentmihalyi, 1990].

However, a system designed to sustain player flow

can sometimes become the very factor that disrupts immersion. In *Donkey Kong Country Returns* (2010), when a player fails multiple times in the same level, the system introduces a helper character named “Super Kong” that automatically clears obstacles, allowing the player to complete the stage without further input. While this design intends to reduce frustration and prevent players from quitting, it paradoxically undermines the psychological foundation of flow. As Chen [Chen, 2007] explains in *Flow in Games*, the essence of flow lies in the “merging of action and awareness” and the “loss of self-consciousness.” When the system substitutes the player’s action, this crucial action–feedback loop is interrupted; the player shifts from being an active participant to a passive observer, losing the sense of mastery and growth that arises from effort. According to Csikszentmihalyi’s theory of flow [Csikszentmihalyi, 1990], the state of optimal experience emerges when players perceive increasing control and skill through challenge. By bypassing this process, “Super Kong” replaces accomplishment with dependency, leading to a loss of self-efficacy and a rupture in immersion. Thus, while such explicit assistance reduces functional difficulty, it simultaneously destroys the continuity and autonomy necessary for sustaining flow.

To avoid the loss of immersion caused by explicit assistance, the *Resident Evil* series has undergone several iterations and optimizations in its DDA systems. In *Resident Evil Village* (2021) and previous installments, Capcom introduced an internal balancing mechanism based on dynamic scoring, which adjusts the overall game difficulty according to player performance during combat. The system calculates a dynamic score based on hit accuracy, damage taken, death retries, and resource consumption, and uses it to modify enemy strength, reaction speed, and item drop rates. However, because the modules controlling enemy strength and resource drops operate independently, the system occasionally produces overlapping or delayed adjustments, causing excessive fluctuations in difficulty and disrupting the rhythm of play.

Building upon this foundation, *Resident Evil 4 Re-make* (2023) implemented a more advanced and integrated version of system-driven Positive DDA. The dynamic scoring system was further divided into two key dimensions: “Action Points” and “Item Points.” The system not only evaluates the player’s combat accuracy and reflexes but also takes into account resource management and consumption efficiency. These two scores combine into a total dynamic score, which determines the player’s current Rank level. Rank variation affects not only player and enemy damage co-

GAME RANK	Player Damage Rate	Enemy Damage Rate	Wince Rate	Break Rate	Stopping Rate	Knife Use Rate
0	0.50	1.30	2.30	1.20	1.20	0.50
1	0.60	1.80	2.00	1.20	1.20	0.70
2	0.70	1.40	1.50	1.10	1.10	0.80
3	0.80	1.20	1.20	1.10	1.10	0.90
4	0.90	1.10	1.10	1.00	1.00	1.00
5	1.00	1.00	1.00	1.00	1.00	1.00
6	1.20	0.90	0.90	1.00	1.00	1.10
7	1.40	0.80	0.80	0.90	0.90	1.20
8	1.60	0.75	0.70	0.85	0.85	1.25
9	1.80	0.70	0.60	0.80	0.80	1.30
10	2.00	0.65	0.55	0.75	0.75	1.35
11	2.20	0.60	0.50	0.70	0.70	1.40

Figure 3: Dynamic Adjustment Parameters in the Rank System of *Resident Evil 4 Remake*

efficients but also multiple combat-related parameters such as Wince Rate, Break Rate, Stopping Rate, and Knife Use Rate. This multidimensional adjustment allows the system to influence combat flow in finer increments, making difficulty changes less about numeric scaling and more about subtle feedback embedded in every shot, dodge, and resource decision. Meanwhile, the remake also redesigned the checkpoint logic, inheriting Action Points from the previous save while reducing retry penalties. This prevents sudden drops in difficulty after minor failures, keeping the overall pacing smooth and consistent. Compared to *Village*'s linear scoring model, *Resident Evil 4 Remake* achieves significantly greater precision and continuity in dynamic balancing through a more refined and fluid multidimensional adjustment system.

The widespread success of *Resident Evil 4 Remake* demonstrates the excellence of its dynamic difficulty system. System-driven Positive DDA, with its high level of automation and real-time feedback, dynamically maintains the balance between challenge and ability without player intervention, embodying the practical application of flow theory in modern game design. Its key strength lies in quantifying player performance indicators (such as accuracy, damage taken, and resource consumption) to instantly recalibrate gameplay variables, ensuring that difficulty changes feel natural and continuous. This prevents players from falling out of the flow channel during repeated failure or extended success. Compared to previous installments, *Resident Evil 4 Remake* transforms system-driven DDA from a simple automatic adjustment mechanism into a "dynamic guidance system" that continuously and subtly tunes feedback to keep the player's psychological state in the optimal "edge of challenge." This natural rhythm allows players to remain focused on their actions rather than on system intervention, thus extending the duration of the flow experience.

The effectiveness of system-driven Positive DDA depends on its balance between precision and subtlety.

Adjustment precision reflects the system's ability to accurately model the player's true skill level and respond appropriately based on behavioral data, ensuring that challenge matches performance. Adjustment subtlety, meanwhile, determines whether the system can modify difficulty without being noticed, preserving a seamless and natural experience. If the adjustment becomes too explicit or reacts too strongly, players become aware of the system's intervention, breaking immersion. Hence, a truly successful system-driven DDA does not make players notice the changes—it allows them to remain, unconsciously, within the equilibrium of flow. This dynamic balance of high precision and high subtlety enables *Resident Evil 4 Remake* not only to achieve challenge-ability alignment on a systemic level but also to maintain a sustained sense of tension and control on a psychological level, representing the most mature evolution of system-driven DDA in modern game design.

4.2 System-driven Negative DDA

System-driven Negative Dynamic Difficulty Adjustment (DDA) is another form of system-controlled adaptive mechanism. Unlike its positive counterpart, it increases difficulty when the player performs poorly rather than lowering it. The system monitors player failures and deliberately strengthens enemies, reduces resources, or weakens the player's capabilities to create additional tension and pressure. The goal is not to alleviate frustration but to heighten intensity, rewarding persistence and mastery through sustained high-stakes gameplay.

Under this DDA mode, the game system continuously applies a stimulus that makes players feel danger and pain. Moreover, when players continue the same behavior, that is, when they fail, the stimulus intensifies further until the players take an action that is the opposite of the previous one. This is clearly a form of positive punishment. It is like walking blindfolded through a thorn bush. Since players are mostly subjected to harmful stimuli for most of the time, they maintain a state of tension during the gameplay. This emotional state amplifies the game's challenge from an experiential perspective. After all, when players' minds are filled with concerns about enemy ambushes and fears of cliff terrains, any originally evenly matched battle will seem extremely challenging.

This form of DDA is rare in mainstream games and most commonly found in Souls-like titles known for their punishing difficulty. One of the most representative examples is the death penalty system: when a player dies, most enemies respawn, and the player's "souls" (experience points) are dropped at the death lo-

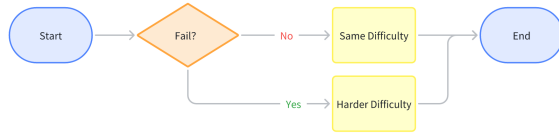


Figure 4: System-driven Negative DDA Loop

cation. If the player fails to retrieve them on the next attempt, all progress is permanently lost. What was once a single challenge now reactivates after each failure, forcing players to confront the same threats multiple times. Combined with the overall difficulty of these games, this system compels players to plan each run carefully, analyze enemy behavior, and refine their strategy. Upon overcoming such obstacles, players experience a powerful sense of achievement, entering a state of flow built on the edge of their capability. However, this high-pressure adjustment system does not always yield ideal flow states. When punishment accumulates with each death, the challenge level can exceed the player’s tolerance, shifting the experience from focused engagement to anxiety and frustration. In *Demon’s Souls* (2009), this issue is particularly evident: its “World Tendency” system darkens the world with each death, increasing enemy health and damage, while the player’s maximum HP is halved in soul form. As failures compound, the difficulty curve steepens beyond manageable limits, pushing players out of the flow channel and into anxiety. The game thus risks creating a “failure spiral,” where frustration overwhelms engagement.

In summary, while negative DDA can amplify tension and deepen immersion for highly skilled players, it more easily drives others out of flow. Because the system increases difficulty after failure rather than easing it, players often face challenges that exceed their capability, leading to anxiety rather than mastery. In essence, by emphasizing punishment to heighten intensity, the system risks breaking the delicate balance between challenge and ability that defines optimal flow.

4.3 Player-driven Implicit DDA

Player-driven Implicit Dynamic Difficulty Adjustment (DDA) refers to a mechanism in which players actively influence the level of challenge through their own actions during gameplay. Unlike system-driven automatic adjustments, this type of DDA emphasizes player autonomy: difficulty variations are not directly controlled by algorithms but emerge naturally from player decisions in areas such as resource management, character progression, or strategic play, thereby indirectly increasing or decreasing the level of challenge. These

mechanisms are often seamlessly embedded within the core gameplay loop, allowing difficulty adjustment to merge with the play experience itself. As a result, they provide freedom and flexibility while preserving immersion and the continuity of Flow.

In *flow* (2006), this concept is expressed in its purest form. Players can choose to dive deeper into the ocean to confront stronger enemies or remain in shallower layers for an easier experience. Difficulty adjustment is directly integrated into the act of exploration, transforming “choosing a challenge” into a natural behavior rather than a menu-based option. However, in practice, most players rarely choose to ascend to the upper layers to reduce difficulty. This phenomenon reflects a universal psychological tendency: people are more inclined to pursue growth and mastery than to retreat or lower their goals. This behavior can be interpreted through multiple psychological frameworks. According to Csikszentmihalyi’s theory of *Flow* [Csikszentmihalyi, 1990], individuals are most likely to experience optimal engagement when challenge and ability are balanced and progress is perceptible. The *Loss Aversion Theory* [Kahneman and Tversky, 1979] suggests that the pain of regression outweighs the pleasure of advancement. *Cognitive Dissonance Theory* [Festinger, 1957] explains that individuals strive to maintain internal consistency and are reluctant to admit their own limitations. Meanwhile, *Self-efficacy Theory* [Bandura, 1977] posits that people are motivated to preserve positive beliefs about their competence; thus, voluntarily lowering challenge is often perceived as a sign of failure or loss of ability. Collectively, these mechanisms drive players to continue confronting higher challenges, maintaining the balance of the flow channel through self-driven persistence.

In contrast, the *Dark Souls* series applies this principle in a more complex and mature way. The game does not automatically adjust difficulty according to player performance; instead, variations in challenge emerge entirely from player choices. Players can repeatedly defeat enemies (grinding) to accumulate “Souls” and strengthen their character, thus reducing the subsequent difficulty. Alternatively, players may choose the sorcerer build to avoid many dangerous close-range combat encounters. They may also summon NPCs or other players for assistance. Conversely, if players reject help, face bosses alone, or impose self-restrictions (such as completing the game without leveling up, wearing armor, or using only low-tier weapons), the difficulty rises significantly. The system never explicitly indicates these changes—the difficulty curve is created and managed entirely by the player. This form of *self-shaped difficulty* grants players a high degree of free-



Figure 5: Magic or Sword

dom while reinforcing their sense of control and accomplishment. Through cycles of failure and mastery, players learn enemy patterns, refine tactics, and improve their execution, effectively “making themselves stronger” rather than “making the game easier.” This design fulfills the goal of DDA—balancing challenge and ability—but achieves it entirely through self-driven progression, transforming system regulation into psychological self-regulation.

Roguelike games further systematize and iterate upon this growth-driven implicit adjustment model. In *Hades* (2020), for example, the “Mirror of Night” system allows players to gain permanent upgrades after each failed run, thereby lowering future difficulty, while the “Heat” system enables them to voluntarily add modifiers that enhance enemy strength and restrictions in exchange for greater rewards. This *bidirectional implicit adjustment* reflects both player growth and voluntary challenge escalation, maintaining dynamic balance and Flow continuity across repeated cycles of death and rebirth. Players experience satisfaction through gradual empowerment and, simultaneously, can reintroduce difficulty to sustain engagement, forming a continuous “challenge–ability–feedback” loop of motivation and control.

Overall, player-driven implicit DDA integrates difficulty modulation directly into the core gameplay, allowing players to unconsciously calibrate challenge intensity through natural interaction. When players engage in actions such as resource allocation, character progression, or strategic decision-making, they are effectively shaping difficulty in accordance with their own competence and preferences. The key strength

of this approach lies in its seamlessness—difficulty adjustment is not presented as an explicit mechanic but as an organic part of gameplay, preserving immersion and the continuity of Flow. However, this implicit autonomy also introduces uncertainty: without real-time system monitoring and feedback, difficulty changes depend largely on player decision-making, resulting in lower precision and consistency compared to system-driven automatic adjustments.

4.4 Player-driven Explicit DDA

Player-driven Explicit Dynamic Difficulty Adjustment (DDA) refers to a mechanism that allows players to directly and visibly adjust the intensity of game challenges. Through menu options, mode selections, or accessibility features, players can actively modify gameplay parameters to shape their experience according to personal preferences. Unlike implicit player-driven DDA, this form of adjustment is entirely transparent and controllable. Players are consciously aware that they are intervening in the system, which enhances their sense of autonomy but may also disrupt the game’s natural rhythm and immersion.

In *God of War* (2018), the game offers four difficulty tiers ranging from “Give Me a Story” to “Give Me God of War.” Although this system is fundamentally a static difficulty structure, players can freely switch between most tiers mid-level, achieving a limited form of “dynamic adjustment.” This flexibility allows players to adapt the challenge level to their current state or mood, improving inclusivity and accessibility. However, from the perspective of Flow theory, such explicit adjustments do not effectively sustain player Flow. As Chen [Chen, 2007] notes in *Flow in Games*, the essence of Flow lies in the “merging of action and awareness” and the “loss of self-consciousness.” When players are immersed in the feedback loop of challenge and response, their attention is entirely focused on the ongoing action. Explicit difficulty changes, however, force them to shift from “awareness-in-action” to “reflection-on-action,” breaking the seamless continuity of experience. This interruption causes players to step outside the game experience and re-enter a meta-cognitive state, becoming aware of their control over the system and thereby disrupting the natural feedback loop essential for sustaining Flow.

In contrast, *The Last of Us* series demonstrates a more sophisticated and inclusive use of explicit DDA. The game not only provides multiple overall difficulty levels but also allows players to individually adjust parameters such as combat intensity, stealth detection, resource scarcity, and enemy perception. It also includes accessi-



Figure 6: Various Accessibility Options in The Last of Us Series

bility options such as aim assist, auto-pickup, and navigation aids, which significantly expand playability for a broader range of players. These fine-grained adjustments enable players of different ability levels to engage with the narrative at their own pace. Notably, the series also incorporates a system-driven Positive DDA that dynamically balances combat rhythm and player performance. Therefore, the player-driven explicit DDA functions less as a primary difficulty regulation mechanism and more as a “fallback system” designed to support players who continue to struggle despite adaptive system balancing.

Thus, explicit DDA represents a necessary design compromise. From the Flow perspective, it sacrifices experiential continuity and naturalness; yet from the standpoint of accessibility and emotional engagement, it ensures that a greater number of players can fully experience the game’s narrative and thematic core. As Chen emphasizes in his conclusion, the ideal implementation of DDA should “embed choices into gameplay,” allowing players to unconsciously self-regulate challenge through play rather than making rational adjustments via external menus [Chen, 2007]. Explicit DDA, therefore, exists as a pragmatic concession—a trade-off between the purity of Flow and the inclusivity of experience, exchanging a degree of immersion for the completeness and universality of player engagement.

5 Discussion

Overall, different types of Dynamic Difficulty Adjustment (DDA) mechanisms play distinct roles in shaping the player experience. System-driven DDA, by continuously monitoring player performance through algorithms, achieves high precision and consistency in dynamically balancing challenge and ability, serving as a core mechanism for maintaining flow. In contrast, player-driven DDA grants players direct control, allowing them to shape personalized experiences through

strategic choices or active adjustments, thereby enhancing their sense of freedom and engagement. However, even within the same type of DDA, variations in implementation and adjustment strategy can significantly affect players’ emotional responses and flow states.

A successful game often does not rely on a single type of DDA, but rather combines multiple forms to achieve broader adaptability. As demonstrated by The Last of Us series, integrating system-driven and player-driven DDA allows the game to balance challenge with accessibility, ensuring that players of varying skill levels can remain engaged and experience the full narrative at their own pace. Thus, a truly exceptional game is not defined merely by the sophistication of its DDA system, but by its ability to employ multi-layered dynamic adjustments—enabling every player to find their own “flow channel” and unique sense of enjoyment within the experience.

6 Future Work

Firstly, the newly introduced *Auto-Sort Inventory* feature in *Resident Evil 4 Remake* (2023) offers a new perspective on the boundaries of Dynamic Difficulty Adjustment (DDA). Although this function does not directly affect enemy strength or combat difficulty, it indirectly reduces the game’s overall complexity by simplifying resource management and lowering the time cost and operational pressure associated with manual inventory organization. Formally, this mechanism belongs to a “Quality of Life” feature; however, its psychological effect—relieving cognitive and operational stress—resembles the goal of DDA in maintaining Flow through adaptive adjustment. It is worth noting that once players become accustomed to this feature, they often develop a strong dependence on it, thereby weakening the strategic experience originally built around manually organizing the inventory. Whether this feature should be classified as a form of DDA remains open to discussion, yet the phenomenon it creates may provide valuable insights for future research on DDA.

Secondly, future studies should expand their focus to a broader range of game genres. Many categories, such as puzzle games, rarely implement DDA systems, possibly due to their smaller audiences and fixed pacing. Investigating whether such genres could benefit from adaptive adjustment mechanisms to enhance playability and immersion may reveal new opportunities for design innovation.

Lastly, different types of DDA may be more suitable for specific game structures. For example, system-driven positive DDA may align better with linear games lacking character progression systems, whereas player-

driven mechanisms tend to fit open-world or role-playing games with self-regulating progression loops. Future research could further analyze how various forms of DDA correspond to different game genres, player motivations, and narrative pacing, thereby deepening our understanding of “dynamic design” as both a technical and experiential framework.

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